IRD

IRD is a cancer drug treatment. It is used to treat a type of blood cancer called myeloma.

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What is IRD?

IRD is a combination of cancer drugs. It is used to treat a type of blood cancer called myeloma.

IRD includes the following drugs:

- I - ixazomib (NinlaroÒ) is a type of targeted therapy drug called a growth or proteasome inhibitor.
- R – lenalidomide (Revlimid®) is a type of cancer drug called an immune system modulator.
- D – dexamethasone is a steroid.

It is best to read this information with our general information about cancer drug treatments and the type of cancer you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree (consent) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the electronic Medicines Compendium (eMC) website, which has patient information leaflets (PIL) for individual drugs.

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How IRD is given

You take ixazomib and lenalidomide as capsules and dexamethasone as tablets. This means you can take IRD at home.

During your course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your treatment, you will have regular blood tests. This is to check that it is safe for you to continue treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your cancer drugs.

Your course of treatment

Your cancer team will discuss your treatment plan with you. They may give you a copy of the plan to take home.

You usually have a course of several cycles of treatment over a few months. Each cycle of IRD usually takes 28 days.

- You take ixazomib once a week, on day 1, 8 and 15 of the cycle.
- You take lenalidomide every day, on day 1 to 21 of the cycle.
- You take dexamethasone once a week on day 1, 8, 15 and 22 of the cycle.

Your cancer team will give you the capsules and tablets to take home. They may give you capsules or tablets of different strengths. Always take IRD exactly as explained. This is important to make sure it works as well as possible for you.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take all your capsules or tablets exactly as they tell you to.

Taking your capsules and tablets

Swallow the capsules and tablets whole with a glass of water. Do not chew, open, break or crush them.

- Take ixazomib capsules 1 hour before food or 2 hours after food. If you forget to take a dose, take it as soon as you realise. But if it is less than 3 days until the next dose, do not take the missed dose.
- You can take lenalidomide with or without food. Try to take them at the same time each day. If you forget to take a dose, take it as soon as you realise. But if it is less than 12 hours until the next dose, do not take the missed capsules.

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- Take dexamethasone with or just after breakfast. If you forget to take dexamethasone, take the missed dose as soon as possible in the same day. If it has been a full day, contact the 24-hour number for advice.

If you are sick (vomit) just after taking any of these drugs, do not take an extra dose. Take your next dose at the usual time. If you are not sure what to do about a missed dose, contact the hospital on the 24-hour number for advice. Never take a double dose to make up for one you have missed.

Other things to remember about your capsules and tablets:

- Wash your hands after taking them.
- Other people should avoid direct contact with them. If possible, you should handle the capsules yourself. If you need help, the person helping you should wear disposable gloves. If pregnant, the person should not touch or handle them at all.
- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
- Do not throw away unused capsules or tablets. Return them to your cancer team at the hospital.

Preventing pregnancy while taking this treatment

You must not become pregnant or make someone pregnant while taking ixazomib and lenalidomide. This is because it may cause severe abnormalities in developing babies.

You will take part in a pregnancy prevention programme during treatment. Even if you do not think you are at risk of getting pregnant or making someone pregnant, you will still be included in this programme.

Your doctor, nurse or pharmacist will give you information about the programme, and the risks of ixazomib and lenalidomide and pregnancy. They will ask you to sign a consent form to show that you understand the information.

Pregnancy tests and contraception

If you are of childbearing age and able to get pregnant, you will have a pregnancy test before starting this treatment. The pregnancy test will be repeated before each cycle, and 4 weeks after treatment finishes.

You must use an effective form of contraception, such as an implant, injection or the progesterone-only pill.

If using the pill, you must also use a barrier method such as a condom. Ixazomib taken with dexamethasone may stop the pill from working.

You should not use the combined oral contraceptive pill. It increases your risk of developing blood clots.

You must use contraception to prevent pregnancy:

- for 4 weeks before treatment

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- during treatment
- for 90 days after treatment finishes.

If you think you may be pregnant at any time during your treatment, contact your doctor, nurse or pharmacist straight away.

Contraception to prevent making someone pregnant

Ixazomib and lenalidomide can pass into the semen that carries sperm. During treatment and for 90 days after treatment finishes, you must not:

- make someone pregnant
- donate sperm.

You can protect partners by using a condom during sex. During treatment and for a week after treatment finishes, you must use a condom during sex with anyone who is:

- pregnant
- of childbearing age and able to get pregnant.

If a partner thinks they might be pregnant during your treatment, contact your doctor, nurse or pharmacist straight away.

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About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects while treatment is being given

Some people may have side effects while they are having this treatment or shortly after they have it.

Allergic reaction

This treatment might cause an allergic reaction. But this is not common. Signs of a reaction can include:

- a skin rash or itching
- feeling breathless or wheezy
- swelling of your face, mouth or throat.

If you feel unwell or have any of these signs, contact the hospital straight away on the 24-hour number. Do not take any more of this treatment until you have checked with them.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

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Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an infection. A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. Your doctor may give you antibiotics and other drugs to take during this treatment. These may help reduce your risk of getting an infection.

But it is still important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

You may be given a drug called G-CSF. This encourages the body to make more white blood cells. You have it as a small injection under the skin.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds

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- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Diarrhoea

This treatment may cause diarrhoea. Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Constipation

This treatment can cause constipation. Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- Eat high-fibre foods, such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Feeling sick

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

If you feel sick, take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

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Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called peripheral neuropathy. You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Skin changes

This treatment can affect your skin. It may cause a rash, which may be itchy. Always tell your doctor, nurse or pharmacist about any skin changes. They can give you advice and may prescribe creams or medicines to help.

Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Muscle or joint pain

You may get pain in your muscles or joints for a few days after treatment. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

Tell them if the pain does not get better. Having warm baths and resting regularly may help.

Eye problems

Your eyes may feel dry or sore. Your doctor, nurse or pharmacist can give you advice and eye drops to help with this.

Contact the hospital on the 24-hour number if:

- your eyes get red and inflamed (conjunctivitis)
- you have pain in your eyes
- you notice any change in your vision.

Feeling tired

Feeling tired is a common side effect of this treatment. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

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Loss of appetite

This treatment can affect your appetite. Don’t worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Effects on the heart

This treatment can affect how the heart works. You may have tests to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing heart problems, your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Raised blood sugar levels

This treatment can raise your blood sugar levels and increase the risk of diabetes.

Signs of raised blood sugar include:

- feeling thirsty
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.

Tell your doctor or nurse if you have these symptoms. They can do tests to check your blood sugar level and give you advice.

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If you already have diabetes, your blood sugars may be higher than usual. You may need to check them more often, or make changes to your diabetic treatment. Your diabetic doctor or nurse will talk to you about how to manage this.

Mood changes

This treatment can affect your mood. Tell your doctor, nurse or pharmacist if you have:

- feelings of anxiety or restlessness
- mood swings (moods that go up and down)
- low mood or depression.

Blood clot risk

This treatment can increase the risk of a blood clot. Your cancer team will give you drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call 999 if you have:

- chest pain
- difficulty breathing.

Changes to blood pressure

This treatment may affect your blood pressure. Tell your doctor or nurse if you have ever had any problems with your blood pressure. Your nurse will check your blood pressure regularly during treatment. Let them know if you feel dizzy or have any headaches.

Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment. TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or rasburicase through a drip.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

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Serious skin changes

Rarely, this treatment can cause a serious skin reaction that needs to be treated immediately in hospital.

Do not take any more of this treatment and contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a skin rash that is spreading
- blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain
- sores on your lips or in your mouth.

Effects on the brain

Rarely, this treatment causes a brain condition that can be serious. You can make a full recovery from this. But it must be diagnosed and treated quickly.

This condition can cause:

- a headache that does not get better
- drowsiness or confusion
- changes in eyesight
- fits (seizures).

If you have any of these symptoms, it is important to either:

- contact the hospital straight away on the 24-hour number
- go to the hospital straight away.

You should not drive yourself to hospital.

Inflammation of the spinal cord

This treatment may cause an inflammation of the spinal cord. This is rare. If you have any of these symptoms, contact the hospital straight away on the 24-hour number:

- weakness in the arms or legs
- sharp pain in the back that moves to the arms and legs
- burning, tingling, numbness or loss of feeling in the legs
- changes to the bowel, such as needing to poo more often, leaking poo, or constipation
- changes to the bladder, such as needing to pee more often, leaking pee, or finding it difficult to pee.

Change in periods

If you have periods, you may find they become irregular, lighter, or stop during this treatment. This does not always mean you are unable to get pregnant. Even if your periods stop or are irregular when

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taking this treatment, you still need to use contraception to prevent a pregnancy. Your doctor, nurse or pharmacist can give you more information about this.

Other important information

Donating blood

You must not give blood during this treatment, or for 1 week after finishing lenalidomide treatment.

Hepatitis B reactivation

Tell your doctor if you have ever had hepatitis B (a liver infection). This treatment can make it active again. Your doctor, nurse or pharmacist will talk to you about this and will test you for it before you start your treatment.

Shingles

You may be more likely to develop a viral infection called shingles during this treatment. Your doctor may give you medicine to help prevent shingles.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- herbal drugs and complementary or homeopathic therapies
- recreational drugs – for example, cannabis.

Lactose

This treatment may contain lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and coronavirus (covid). They may also recommend other vaccines, such as Shingrix® for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

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It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Breastfeeding

You are advised not to breastfeed while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can get pregnant or make someone pregnant. If you are worried about this, it is important to talk with your doctor before you start treatment.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Visit the electronic Medicines Compendium (eMC) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

Cancer Information Development team

Author

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Our cancer information meets the PIF TICK quality mark.

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This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

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Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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